

another view of integer multiplication

I want to explore the natural numbers and integers via a formalist philosophy that is not set-theoretical, adding operations and relations and seeing what objects we came up with. I will aim towards representing $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ and polynomial rings of these; the formalism of what happens next is well codified in standard undergraduate. The aim is not to get there directly, but to explore on our way there.

The main relations of interest will be:

- existence: imposing the existence of an element, and then if any operation is added imposing (formal) completion.
- identification: imposing $x * x = x$
- vitalization: imposing the existence of an inverse pivoted around an identity. **Why is it natural to virtualize when there is an identity; delete this when you figured it out**
- element-wise absorption law: given two separate elements x, y is $xy = x$ or $xy = y$

We shall always assume:

- associativity: This is because natural numbers and functions can both have their multiplication and compositing broken down into “blocks” (see **ref:HERE** for the intuition). As one of the ultimate goals here is to have a better intuition on the primes ideals of $\mathbb{Z}$ and $k[x]$, if we don’t impose associativity we are asking the “wrong” question.
- commutativity: lack of associativity means that we can’t break down our problem and would have to examine each pair. But with associativity we can think of a sequence of operations being performed in order (where we could “simplify” not needing to “collapse” in sequence but can collapse any two in the chain until we get the answer) and so it is well-defined to write:

$$a_1 * a_2 * \cdots * a_n$$

Now, lack of commutativity means that order matters when combining two elements. Commutativity represents an “invariance” to direction. If we have associativity and commutativity, then we can think of it as combining operations that is not sensitive to order. Again, if order mattered we would be dealing with something elem so we impose this too.

- distributivity: When there is more than one binary operation, they can either act “independently” from each other giving multiple new elements using formal combinations or (what is naturally more interesting) we want them to meaningfully interact. For an example of what happens if we are not careful, Define $+$ and $\cdot$ on $\{0, 1\}$ so that $0 + 0 = 0$ and $1 + 1 = 1$, but $0 \cdot 0 = 1$ and $1 + 1 = 0$.

There are many choices that can be made for combining binary operations depending on what we are modelling. In lattice theory, the absorption law is a natural choice. When function addition and composition are taken on affine linear functions, right distributivity is natural.

If we are concerned with area as we add dimensions, then distributivity is a natural candidate. With commutativity, it suffices to define left-distribution. This imposes the following diagram to commute:

$$\begin{array}{ccc}
 R \times R \times R & \xrightarrow{(\text{id}, +)} & R \times R \\
 \downarrow (\cdot, \cdot) & & \downarrow \cdot \\
 R \times R & \xrightarrow{+} & R
 \end{array}
 \qquad
 \begin{array}{ccc}
 (r, s, t) & \xrightarrow{(\text{id}, +)} & (r, s + t) \\
 \downarrow (\cdot, \cdot) & & \downarrow \cdot \\
 (r \cdot s, r \cdot t) & \xrightarrow{+} & r \cdot s + r \cdot t = r \cdot (s + t)
 \end{array}$$

I will try to recover $\mathbb{N}$ by adding restrictions and see if we get something interesting.

Let's consider free commutative semigroup, so group but without necessarily identity or invertibility but associativity. We keep associativity because order of multiplication can be done "per pair of prime divisors" (see [ref:HERE](#) for the intuition for associativity).

Let's take the free cyclic semigroup generated by a singleton, say ϵ . Then this would give us the "multiplication" structure. We can shorten notation and write ϵ^n , which is well-defined notation because of associativity.

We would like